Expert Systems with Applications

When Feature Engineering Outperforms Deep Learning: Sparsity-Aware Direct Multi-Step Forecasting for Small-Scale Retail Demand

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Date
14 October 2026

This title page is submitted separately for double-anonymized peer review. It is not part of the anonymized manuscript.

Acknowledgments

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of competing interests

The author declares that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Zhanpeng Qin: Conceptualization, Methodology, Software, Formal Analysis, Investigation, Writing - Original Draft, Writing - Review & Editing, Visualization, Project Administration.

Data availability

Raw transaction-level data cannot be shared because of commercial confidentiality. Aggregated daily-level statistics sufficient to reproduce the main experimental results are available from the corresponding author upon reasonable request. Source code is publicly available at `https://github.com/qzpthuhhu/eswa-retail-forecasting`.